

A

Project Report

On

CYBERSECURITY AWARENESS TRAINING PLATFORM

Submitted by

Dhairya Kahar [22405101020009]

as

Partial fulfilment of Semester VI

of Bachelor of Science in Information Technology

for A.Y. 2026-2027

Under the Guidance of

Internal Guide name :

Prof. Ramchandaran

Submitted To

Parul Institute of Computer Application,

Faculty of IT & Computer Science

Parul University



Acknowledgement

The success and final outcome of this project required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of our project. All that we have done is only due to such supervision and assistance and we would not forget to thank them.

I respect and thank **Dr. Priya Swaminarayan (Dean)** and **Dr. Hina Chokshi (Vice-Principal)** for providing us an opportunity to do the project work in **BSCIT Regular/Hons** and giving us all support and guidance which made us complete the project duly. We are extremely thankful to Mam for providing her support and guidance, although she had a busy schedule managing the academic affairs.

We would not forget to remember **Prof. Vipul Gamit (HOD)** for his encouragement and more over for his timely support and guidance till the completion of our project work.

We owe our deep gratitude to our project guide **Prof. Ramchandaran**, who took keen interest on our project work and guided us all along, till the completion of our project work by providing all the necessary information for developing a good system.

I am thankful to and fortunate enough to get constant encouragement, support and guidance from our Parents, all Teaching staff of the BCA/IT Department which helped us in successfully completing our project work. Also, we would like to extend our sincere esteems to all staff in the laboratory for their timely support.

Dhairya Kahar [22405101020009]



PARUL INSTITUTE OF COMPUTER APPLICATION

CERTIFICATE

This is to certify that **Dhairya Kahar** the student of Parul Institute of Computer Application, has satisfactorily completed the project entitled “**CYBERSECURITY AWARENESS TRAINING PLATFORM**” as a part of course curriculum in **BSCIT Regular/Hons semester- VI** for the academic year **2026-2027** under guidance of **Prof. Ramchandaran**.

Enrolment Number: 22405101020009

Quality of work	Grade	Sign of Internal guide
Poor / Average / Good / Excellent	B / B+ / A / A+	

Date of submission: _____

HOD,
Prof. Vipul Gamit

Vise-Principal,
Dr. Hina Chokshi

Dean,
Dr. Priya Swaminarayan

INDEX

Content	Page No.
1. Introduction to Project System	1
2. System Requirement Specification	2
2.1 Introduction to SRS	2
2.2 Hardware Requirement	2
2.3 Software Requirement	3
2.4 System Users	3
2.5 Description of User Role	3
2.6 System Modules	4
2.7 Description of Modules	4
2.8 Timeline Chart	6
3. System Flow Diagram	7
4. Data Flow Diagram (All Levels of DFDs)	8
5. Use Case Diagram	11
1. Use Case	11
2. Activity Diagram	12
3. Class Diagram	13
6. Data Dictionary	14
7. Screenshots of Development Phase -1	16
8. Screenshots of Development Phase -2	18
9. Screenshots of Development Phase -3	20
10. Conclusion	22
11. Future Enhancement	23
12. References	24

ABSTRACT

In the contemporary hyper-connected digital era, the proliferation of online services has triggered an alarming surge in sophisticated cyber threats, including deceptive phishing schemes, social engineering manipulation, credential stuffing, and destructive ransomware campaigns. Extensive global empirical studies consistently demonstrate that over ninety percent of documented security breaches originate from human vulnerabilities, cognitive oversights, and deficient digital hygiene, rather than direct cryptographic or architectural exploitation of underlying server infrastructures. Conventional instructional methodologies rely predominantly on static compliance documents and passive instructional media, resulting in disengagement and acute deficits in practical threat recognition capabilities.

To decisively mitigate this critical human risk factor, the **“Cybersecurity Awareness Training Platform”** was conceptualized, engineered, and deployed as a full-stack, experiential web-based educational ecosystem. Architected upon a robust Python 3.12 Flask framework, SQLAlchemy ORM, and serverless PostgreSQL 16 on Neon Cloud, with a highly responsive Bootstrap 5 interface, the platform delivers eight structured micro-learning curriculum domains. The core technical innovations comprise an experiential Phishing Simulation Sandbox that replicates authentic deceptive email headers, an interactive client-side Password Strength and Shannon Entropy Analyzer estimating GPU brute-force crack timelines, and an expanded 40-question randomized assessment suite delivering automated grading and explanatory pedagogical rationales. The platform provides a measurable, scalable, and production-ready pedagogical asset that elevates institutional cyber resilience through proactive habit formation.

1. Introduction to Project System

1.1 Project Overview & Background

In an era defined by rapid digital transformation, cloud integration, and widespread adoption of online services, cybersecurity has transitioned from an esoteric administrative discipline into an existential personal and organizational imperative. As academic institutions, financial infrastructures, and corporate enterprises migrate operational workflows into distributed web ecosystems, the attack surface exposed to adversaries has expanded exponentially. Sophisticated threat actors increasingly direct their offensive efforts away from fortified network perimeters, opting instead to exploit the most vulnerable link in the security perimeter: human decision-making.

The **Cybersecurity Awareness Training Platform** is engineered as a modern, interactive, full-stack educational web application designed to systematically remediate human-centric security risks. The platform moves beyond theoretical lecturing by delivering an experiential learning laboratory where users analyze real-world threat indicators, observe defensive mathematical concepts in real time, and quantify their cyber defense readiness through automated evaluative feedback.

1.2 Problem Statement & Motivation

Statistical analyses conducted by leading threat intelligence agencies reveal that human error accounts for over 85% to 90% of successful enterprise data breaches. The prevailing methods utilized across corporate and educational institutions rely almost exclusively on passive educational paradigms: lengthy PDF handouts, sporadic compliance circulars, and mandatory recorded webinars. These methods suffer from critical structural deficiencies:

- **Cognitive Disconnect:** Passive reading fails to instill visceral recognition of deceptive cues, such as lookalike domain spoofing or psychological pretexting.
- **Absence of Practical Tools:** Users are instructed to formulate "strong passwords" without mathematical intuition regarding entropy, character space, or brute-force computational velocities.
- **Zero Immediate Verification:** Traditional educational delivery lacks automated, randomized assessment pipelines that pair evaluation with immediate pedagogical rationales.

1.3 Project Purpose & Objectives

The primary purpose of this project is to develop and deploy an accessible, robust, and engaging cybersecurity educational platform that achieves the following concrete technical and educational objectives:

1. Engineer a secure, multi-tier web application using Python 3.12 Flask and SQLAlchemy ORM, incorporating robust cryptographic authentication (PBKDF2/SHA-256).
2. Construct an intuitive client-side **Password Entropy & Strength Analyzer** calculating Shannon information entropy and estimating GPU cluster crack timelines.
3. Implement an experiential **Phishing Simulation Sandbox** allowing learners to safely examine deceptive headers, spoofed senders, and malicious embedded links.
4. Deploy a randomized **40-Question Assessment Engine** with dynamic PostgreSQL persistence, automated scoring algorithms, and instant explanatory feedback.
5. Provide role-based analytical dashboards for learners to monitor completion trajectories, and an administrative control panel for curriculum governance.

2. System Requirement Specification

2.1 Introduction to SRS

The Software Requirement Specification (SRS) establishes a formal, rigorous contract delineating the complete operational, functional, and non-functional specifications governing the Cybersecurity Awareness Training Platform. It specifies the environment, constraints, data structures, and hardware/software boundaries required to achieve production stability, rapid response latency, and uncompromising defensive posture across client and cloud server environments.

2.2 Hardware Requirement

The system is designed with high operational efficiency, ensuring that the client-side presentation renders smoothly even on low-power mobile viewports, while the backend server efficiently handles concurrent sessions with minimal computational overhead.

Table 2.1: Hardware Requirements Specification

Hardware Node	Minimum Specification	Recommended Specification
Client Processor (CPU)	Intel Core i3 / AMD Ryzen 3 / Dual-Core 1.8 GHz	Intel Core i5 / AMD Ryzen 5 / Quad-Core 2.4 GHz or higher
Client Memory (RAM)	2 GB DDR3/DDR4 RAM	4 GB – 8 GB DDR4/DDR5 RAM
Client Storage	50 MB free disk space (browser cache)	500 MB free disk space
Client Display Resolution	1024 × 768 pixels (Standard Tablet / Desktop)	1920 × 1080 pixels (Full HD) / Responsive Mobile Viewport
Client Network Adapter	Standard Wi-Fi / Ethernet (1 Mbps connection)	Broadband Internet connection (> 10 Mbps)
Server Processor (Cloud/Host)	Single Virtual CPU (vCPU @ 2.0 GHz)	Dual vCPU @ 2.5 GHz or higher
Server Memory (RAM)	512 MB RAM (Container Minimal Footprint)	1 GB – 2 GB RAM (Supporting Concurrent Pooling)
Server Disk Storage	1 GB SSD storage for binaries and assets	10 GB High-Throughput NVMe SSD Storage
Database Storage Node	Neon Serverless Compute with 0.25 Compute Units	Auto-scaling Neon Cloud PostgreSQL with 10 GB storage

2.3 Software Requirement

The software ecosystem employs robust, open-source technologies adhering to modern enterprise web design and defensive software engineering standards.

Table 2.2: Software Environment Specifications

Software Layer	Specification, Libraries & Deployment Tools
Operating System	Development: Windows 11 Pro 64-bit; Production: Ubuntu Linux 22.04 LTS (Containerized)
Programming Language	Python 3.12 (Strict typing, async capabilities, robust standard cryptographic libraries)
Web Framework	Flask 3.0.3 (WSGI micro-framework, Blueprint routing architecture, Jinja2 templating)
Object-Relational Mapping	Flask-SQLAlchemy 3.1.1 / SQLAlchemy 2.0 (Parameterized queries, connection pooling)
Authentication & Session	Flask-Login 0.6.3; Werkzeug 3.0 (PBKDF2/SHA-256 salted password hashing)
Cloud Database	PostgreSQL 16 (Neon Serverless Cloud instance with SSL/TLS encryption)
Local Database Fallback	SQLite3 (Built-in Python driver for zero-dependency local offline development)
Client Frontend Framework	HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3.3, Font Awesome 6 Iconography
WSGI Production Server	Gunicorn 22.0.0 (Pre-fork worker model for production concurrency on Render)
Cloud Hosting Platform	Render Cloud Platform (Automated CI/CD from Git repository)

2.4 System Users

The platform accommodates three primary user profiles:

- **Student / General Learner:** Enrolled users seeking to enhance security awareness, complete learning modules, test password strength, run phishing simulations, take evaluative quizzes, and track completion telemetry.
- **System Administrator:** Privileged user responsible for curriculum governance, question bank authoring/editing, system auditing, and monitoring user performance analytics.
- **Guest / Unauthenticated Visitor:** Can view platform landing page, review course syllabus summaries, and register for a verified account.

2.5 Description of User Role

Role-based access control (RBAC) enforces strict authorization barriers across endpoints:

- **Learner Role:** Authorized to perform read operations on educational content; execute client-side simulation scripts; create and persist quiz attempts; read personal score records and dashboard telemetry. Learners are strictly barred from modifying question records or viewing other users' attempts.
- **Administrator Role:** Possesses superset privileges guarded by the `@admin_required` decorator. Authorized to execute full CRUD (Create, Read, Update, Delete) operations on assessment questions and curriculum topics; monitor system logs; and audit institutional score distributions.

2.6 System Modules

The platform architecture decomposes into seven tightly integrated functional modules operating within a clean Flask Blueprint structure:

1. **User Authentication & Profile Lifecycle Module:** Handles registration, session creation, and credential hashing.
2. **Micro-Learning Security Curriculum Module:** Organizes 8 structured defensive security domains.
3. **Real-Time Password Strength & Entropy Analyzer:** Computes Shannon bit entropy and GPU cracking benchmarks.
4. **Experiential Phishing Simulation Sandbox:** Provides interactive inspection of spoofed email headers and indicators.
5. **40-Question Assessment & Evaluation Engine:** Standardized quiz delivery with dynamic randomization and grading.
6. **Learner Analytics & Progress Dashboard:** Visual progress tracking, completion badges, and score trends.
7. **Administrative Governance & Question Bank CRUD:** Privileged interface for curriculum management.

2.7 Description of Modules

Module 1: User Authentication & Profile Lifecycle (`auth.py`)

This module provides secure onboarding and stateful session control. Registration enforces stringent client-side and server-side validation (minimum 8 characters, confirmation match, unique email check). Passwords undergo salted key derivation using PBKDF2 with SHA-256 iterations via Werkzeug. Plaintext credentials never touch memory or disk. Sessions utilize cryptographically signed cookies with HTTPOnly and SameSite flags, neutralizing session hijacking and script-based credential extraction.

Module 2: Micro-Learning Security Curriculum (`main.py`)

The educational curriculum is partitioned into eight foundational cybersecurity domains stored in relational database entities:

- **Domain 1 – Password Security & Credential Hygiene:** Entropy principles, passphrases, credential stuffing defense.
- **Domain 2 – Phishing Attacks & Social Engineering:** Spear phishing, deceptive URLs, emotional urgency triggers.
- **Domain 3 – Malware & Ransomware Defense:** Trojan delivery, ransomware encryption mechanisms, offline backups.
- **Domain 4 – Safe Web Browsing & HTTPS:** TLS certificate validation, mixed content risks, third-party cookies.
- **Domain 5 – Two-Factor & Multi-Factor Authentication:** TOTP algorithms, SMS intercept vulnerabilities, FIDO2 hardware keys.
- **Domain 6 – Public Wi-Fi Risks & VPN Encryption:** Evil twin APs, packet sniffing, encrypted tunneling protocols.
- **Domain 7 – Mobile & Smart Device Hardening:** App permissions, sideloading risks, remote wipe capabilities.
- **Domain 8 – Legal Framework & Incident Reporting:** Information Technology Act 2000 provisions, 1930 Cyber Helpline.

Module 3: Password Strength & Entropy Analyzer (`main.py` / `JS`)

Provides an interactive mathematical laboratory. When the user types a candidate password, the client-side engine calculates the character pool size (S) and length (L) to derive Shannon information entropy: $H = L \cdot \log_2(S)$. It dynamically maps entropy bits to brute-force crack times assuming an offline cluster calculating 10^{10} hashes/second, categorized into Very Weak (<28 bits), Weak (28–35), Fair (36–59), Strong (60–127), and Military Grade (128+ bits).

Module 4: Experiential Phishing Simulation Sandbox (`main.py`)

Unlike passive text descriptions, this sandbox presents learners with authentic, visual mockups of sophisticated deceptive emails (e.g., fraudulent banking notifications, spoofed CEO wire transfers, cloud account deactivation warnings). The learner inspects the message interface, checks the sender header, and hovers over embedded hyperlinks to observe spoofed target URLs. Upon selecting “Legitimate” or “Phishing”, the engine performs instantaneous heuristic evaluation, displaying red-flag callouts: lookalike domains (e.g., `paypa1.com`), generic greetings, artificial urgency, and misaligned return-path headers.

Module 5: 40-Question Assessment & Evaluation Engine (`quiz.py`)

The evaluative core of the platform is an expanded 40-question randomized assessment suite. Questions are relationally mapped to the 8 curriculum topics (5 standardized questions per topic). Features include:

- **Dynamic Randomization:** Questions are retrieved dynamically from the cloud database, preventing fixed-pattern memorization.
- **Automated Server-Side Scoring:** Form submission validates selected options against the stored correct key, computes raw scores, and derives accurate percentage metrics.
- **Comprehensive Solution Breakdown:** Immediate feedback identifies correct answers alongside comprehensive technical explanations detailing why specific choices are vulnerable.
- **Persistent Scoring Records:** Attempt metadata (user ID, topic ID, score, total, percentage, timestamp) is committed transactionally to the `scores` table.

Module 6: Learner Analytics & Progress Dashboard (`main.py`)

The dashboard aggregates real-time learning metrics for authenticated users. It queries the `topic_progress` and `scores` tables to render:

- **Curriculum Completion Ratio:** Visual percentage progress bar denoting finished topics.
- **Cumulative Quiz Performance:** Mean score percentage across all submitted evaluations.
- **Detailed Assessment History:** Tabular log of past quiz attempts, timestamps, and performance grades (Passed / Retake Recommended).

Module 7: Administrative Governance & Question Bank CRUD (`admin.py`)

Guarded by role-based authorization, this module equips administrators to maintain educational standards:

- **Question Management:** Form-based CRUD interface to draft, preview, update, and remove questions from the live database.
- **Platform Telemetry:** Oversight tables displaying enrolled student counts, attempt frequencies, and aggregate score distributions to identify topics requiring curriculum refinement.

2.8 Timeline Chart

The project followed a disciplined Software Development Life Cycle (SDLC) across 12 calendar weeks spanning the academic term. The timeline below illustrates the planned and executed milestones across all phases:

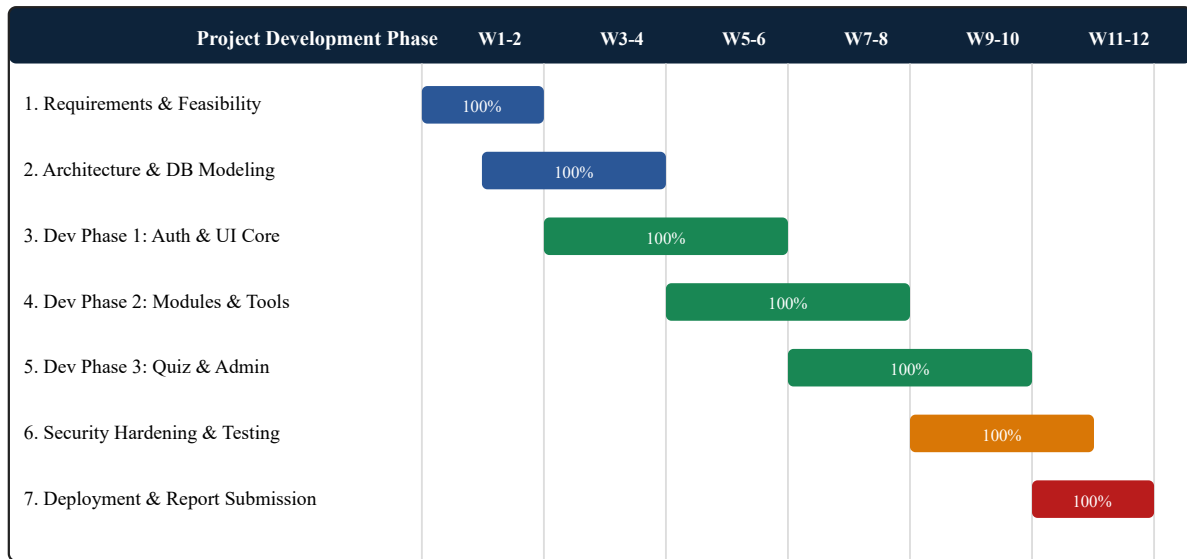


Figure 2.1: Project Development Life Cycle & Execution Timeline Chart

The development schedule successfully observed critical academic milestones: foundational literature and feasibility review (Weeks 1–2), entity relational design and schema migration (Weeks 3–4), Phase-1 UI and cryptographic authentication (Weeks 4–6), Phase-2 interactive phishing/entropy development (Weeks 6–8), Phase-3 dynamic 40-question quiz engine and admin suites (Weeks 8–10), followed by end-to-end security penetration testing, cloud staging on Render, and final documentation (Weeks 11–12).

3. System Flow Diagram

The System Flow Diagram illustrates the holistic runtime execution sequence, decision gates, authentication checkpoints, and navigational pathways traversed by users across the application ecosystem.

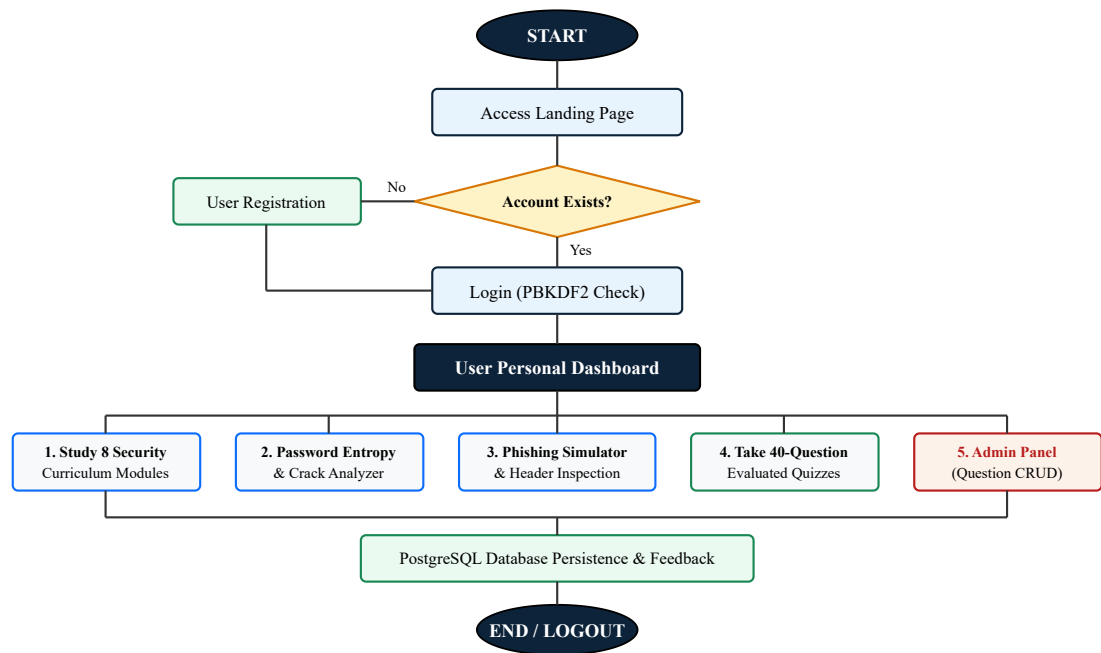


Figure 3.1: Holistic System Flow & User Execution Sequence

The execution flow enforces authentication boundaries before exposing evaluative workflows. Authenticated users access a centralized dashboard offering direct entry to educational modules, cryptographic password testing, experiential phishing scenarios, and formal quizzes. State transitions, attempt metrics, and progress updates commit atomically to PostgreSQL, ensuring synchronization between learning telemetry and dashboard visualizations.

4. Data Flow Diagram (All Levels of DFDs)

Data Flow Diagrams (DFDs) provide a graphical representation of the information pipeline through the system, modeling external entities, operational transformation processes, and underlying datastores without regard to hardware implementation.

4.1 DFD Level 0 – Context Level Diagram

Figure 4.1 depicts the top-level boundary diagram, illustrating all data streams flowing into and out of the primary system process (0.0 Cybersecurity Awareness Platform) across external actor boundaries.



Figure 4.1: Data Flow Diagram Level 0 – Context Level Diagram

In the context diagram, external entities (Student and Administrator) communicate exclusively through formal HTTP/HTTPS request-response cycles with the central platform. The process encapsulates all cryptographic hashing, scoring logic, and database operations behind clean architectural boundaries.

4.2 DFD Level 1 – Subsystem Functional Decomposition

DFD Level 1 decomposes the central system into distinct sub-processes, mapping information exchanges with dedicated relational datastores (D1: users, D2: topics, D3: questions, D4: scores, and D5: topic_progress).

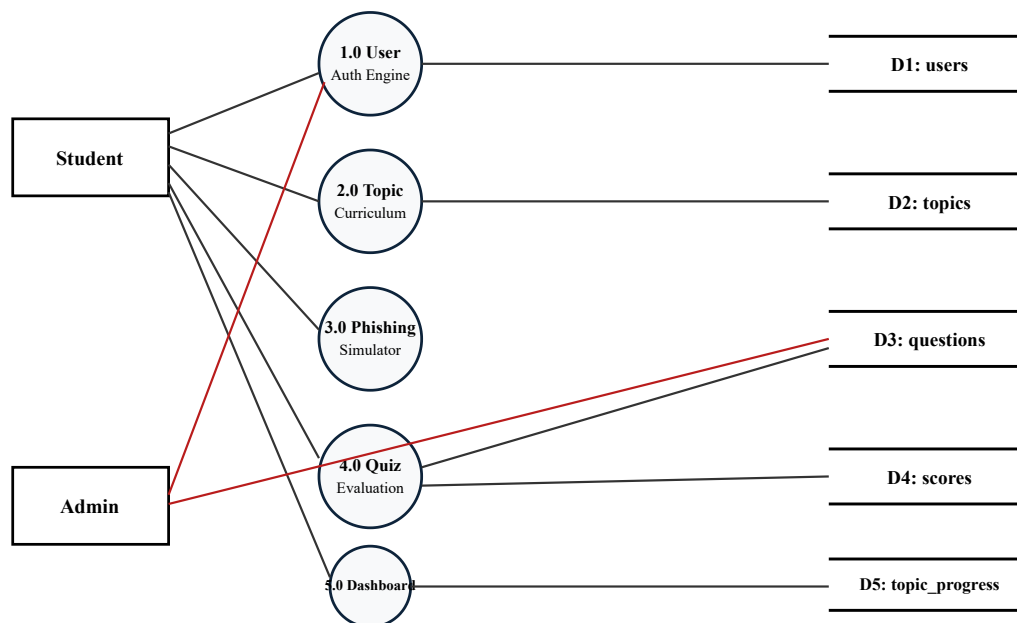


Figure 4.2: Data Flow Diagram Level 1 – Subsystem Functional Decomposition

4.3 DFD Level 2 – Phishing Simulation Subsystem

Figure 4.3 details the internal data flows of Process 3.0 (Phishing Simulation Sandbox).

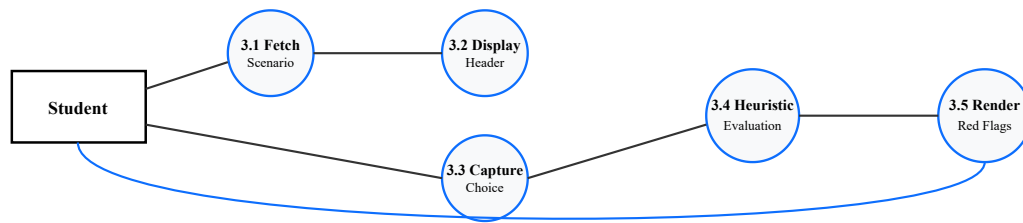


Figure 4.3: DFD Level 2 – Phishing Simulation & Threat Verification Subsystem

4.4 DFD Level 3 – Quiz Assessment & Automated Scoring Pipeline

Figure 4.4 zooms into Process 4.0, depicting the automated evaluation and scoring pipeline.

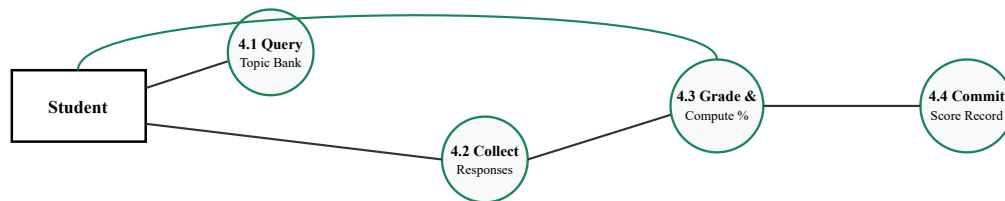


Figure 4.4: DFD Level 3 – Quiz Assessment & Evaluation Pipeline

5. Use Case Diagram

1. Use Case Diagram

The Use Case Diagram defines the operational interactions between system actors (Student and Administrator) and the platform's functional use cases within the application boundary.

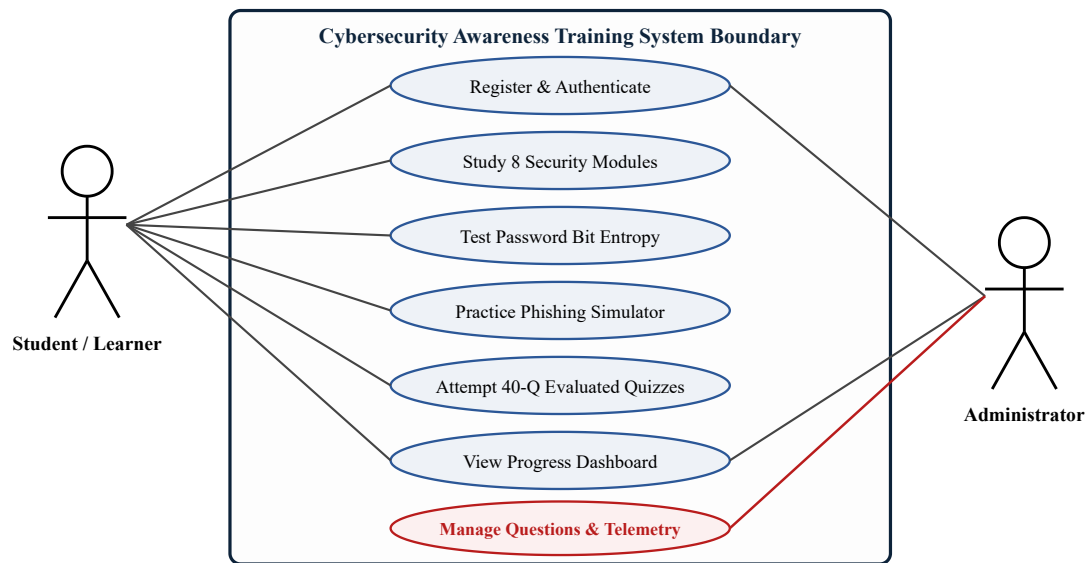


Figure 5.1: UML Use Case Diagram – Actor & System Functional Boundaries

2. Activity Diagram

The Activity Diagram models the procedural workflow, branching logic, and sequential actions undertaken during an active user learning and assessment session.

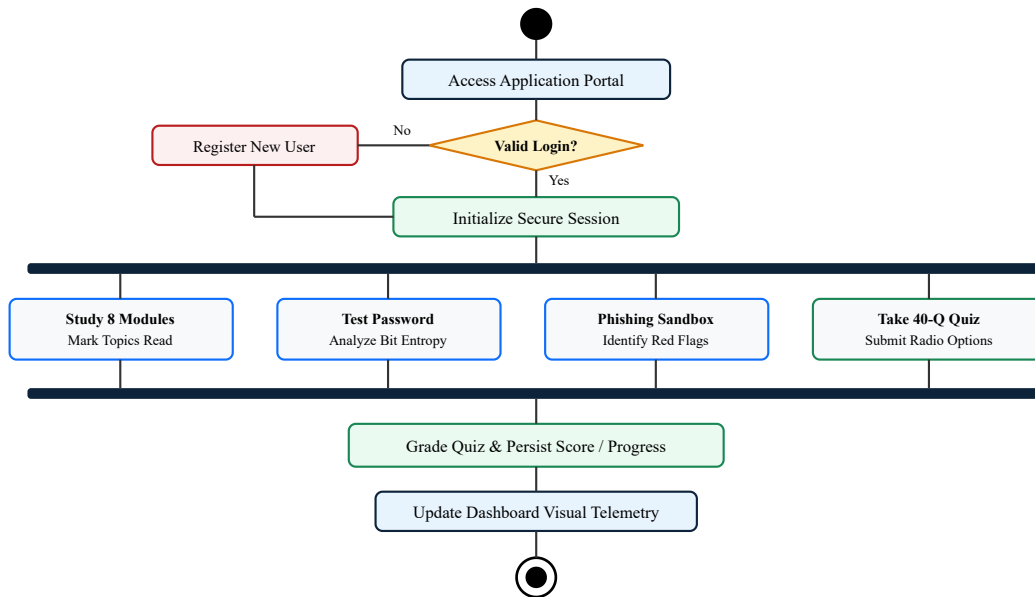


Figure 5.2: UML Activity Diagram – End-to-End User Learning & Assessment Journey

3. Class Diagram

The UML Class Diagram specifies the structural design, model attributes, member functions, access modifiers, and relational associations implemented in the application backend.

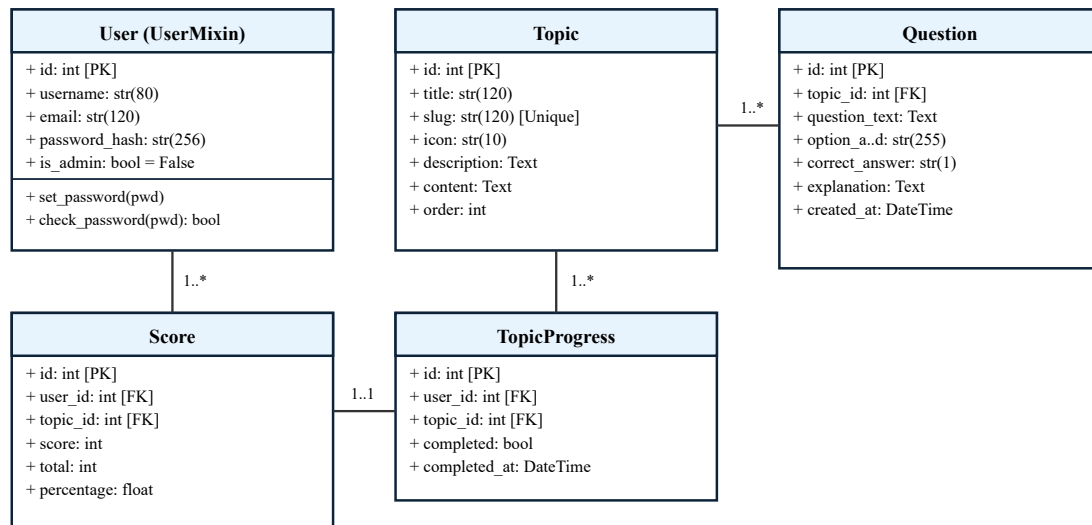


Figure 5.3: UML Class Diagram – Domain Entity Architecture & Service Logic

6. Data Dictionary

The Data Dictionary provides an unambiguous reference describing the relational database schema, detailing attribute names, physical data types, constraints, and operational purposes across all tables.

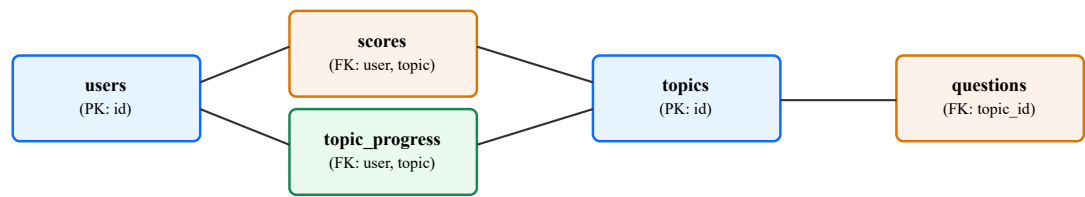


Figure 6.1: Entity-Relationship Diagram (Normalized 3NF Schema)

Table 6.1: Data Dictionary – Table: users

Column Name	Data Type	Constraints	Description & Operational Purpose
id	INTEGER	PK, Auto Increment	Unique system identifier for registered user.
username	VARCHAR(80)	NOT NULL, UNIQUE	Unique handle displayed on dashboard.
email	VARCHAR(120)	NOT NULL, UNIQUE	User email utilized for login credentials.
password_hash	VARCHAR(256)	NOT NULL	PBKDF2/SHA-256 salted cryptographic hash.
is_admin	BOOLEAN	DEFAULT FALSE	Privilege flag guarding administrative endpoints.
created_at	DATETIME	DEFAULT UTC_NOW	Timestamp of initial user account creation.

Table 6.2: Data Dictionary – Table: topics

Column Name	Data Type	Constraints	Description & Operational Purpose
id	INTEGER	PK, Auto Increment	Primary key for curriculum domain.
title	VARCHAR(120)	NOT NULL	Display heading of the security domain.
slug	VARCHAR(120)	NOT NULL, UNIQUE	URL-safe routing identifier.
icon	VARCHAR(10)	NOT NULL	Font Awesome / Unicode visual emblem.
description	TEXT	NOT NULL	Executive summary displayed on dashboard cards.
content	TEXT	NOT NULL	Comprehensive rich HTML instructional body.
order	INTEGER	DEFAULT 0	Display sequence on curriculum listings.

Table 6.3: Data Dictionary – Table: questions

Column Name	Data Type	Constraints	Description & Operational Purpose
id	INTEGER	PK, Auto Increment	Primary key for evaluation question record.
topic_id	INTEGER	FK → topics.id, CASCADE	Foreign key binding question to security topic.
question_text	TEXT	NOT NULL	The assessment stem presented to the student.
option_a	VARCHAR(255)	NOT NULL	First multiple-choice response option.
option_b	VARCHAR(255)	NOT NULL	Second multiple-choice response option.
option_c	VARCHAR(255)	NOT NULL	Third multiple-choice response option.
option_d	VARCHAR(255)	NOT NULL	Fourth multiple-choice response option.
correct_answer	VARCHAR(1)	NOT NULL (A/B/C/D)	Character code matching the single valid answer.
explanation	TEXT	NOT NULL	Pedagogical rationale displayed upon evaluation.
created_at	DATETIME	DEFAULT UTC_NOW	Timestamp of question creation in database.

Table 6.4: Data Dictionary – Table: scores

Column Name	Data Type	Constraints	Description & Operational Purpose
id	INTEGER	PK, Auto Increment	Primary key for quiz submission record.
user_id	INTEGER	FK → users.id, CASCADE	Foreign key referencing the student who took quiz.
topic_id	INTEGER	FK → topics.id, CASCADE	Foreign key referencing evaluated topic.
score	INTEGER	NOT NULL	Total count of correctly answered questions.
total	INTEGER	NOT NULL	Total number of questions in evaluation suite.
percentage	FLOAT	NOT NULL	Calculated achievement percentage (0.0 – 100.0).
taken_at	DATETIME	DEFAULT UTC_NOW	Timestamp when quiz was graded and recorded.

Table 6.5: Data Dictionary – Table: topic_progress

Column Name	Data Type	Constraints	Description & Operational Purpose
id	INTEGER	PK, Auto Increment	Primary key for module progress record.
user_id	INTEGER	FK → users.id, CASCADE	Foreign key referencing enrolled student.
topic_id	INTEGER	FK → topics.id, CASCADE	Foreign key referencing completed topic.
completed	BOOLEAN	DEFAULT FALSE	Boolean tracking reading completion status.
completed_at	DATETIME	NULLABLE	Timestamp when user marked module completed.

7. Screenshots of Development Phase -1

Development Phase -1 focused on the foundational infrastructure: database initialization, user credential lifecycle, cryptographic password validation, and primary authentication interfaces.

UI Mockup 7.1: User Registration & Password Validation<https://cybersecurity-awareness.onrender.com/auth/register>

Create Student Account

Enter institutional credentials to enroll

Username:

Email:

Password:

✓ Strong password meets NIST complexity guidelines

Register Account

Figure 7.1: UI Mockup 7.1 – User Registration & Policy Validation Interface

UI Mockup 7.2: Secure User Authentication & Session Sign-In<https://cybersecurity-awareness.onrender.com/auth/login>

Sign In to CyberPortal

PBKDF2 Cryptographic Verification Active

Registered Email:

Password:

Authenticate & Open Dashboard

Figure 7.2: UI Mockup 7.2 – Secure Login & Session Initialization View

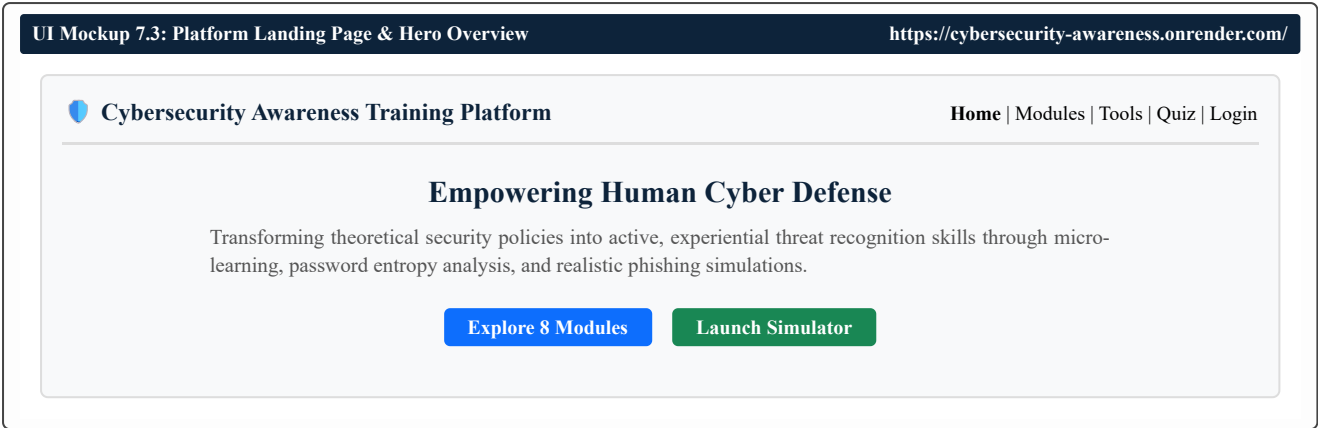


Figure 7.3: UI Mockup 7.3 – Platform Homepage & Hero Overview Interface

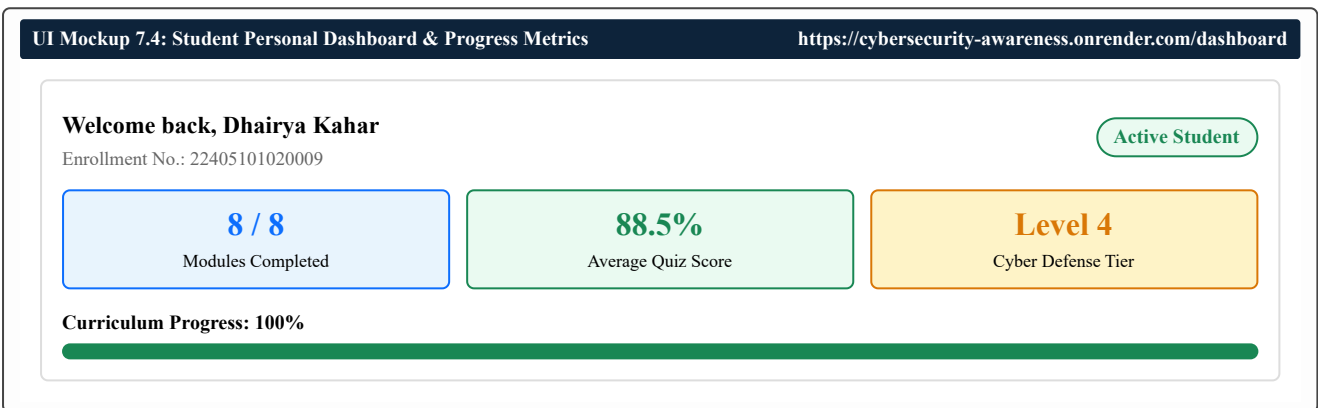


Figure 7.4: UI Mockup 7.4 – Learner Analytics & Personal Dashboard View

8. Screenshots of Development Phase -2

Development Phase -2 focused on core pedagogical delivery: rich micro-learning modules, the real-time client-side Password Entropy Analyzer, and experiential threat sandboxes.

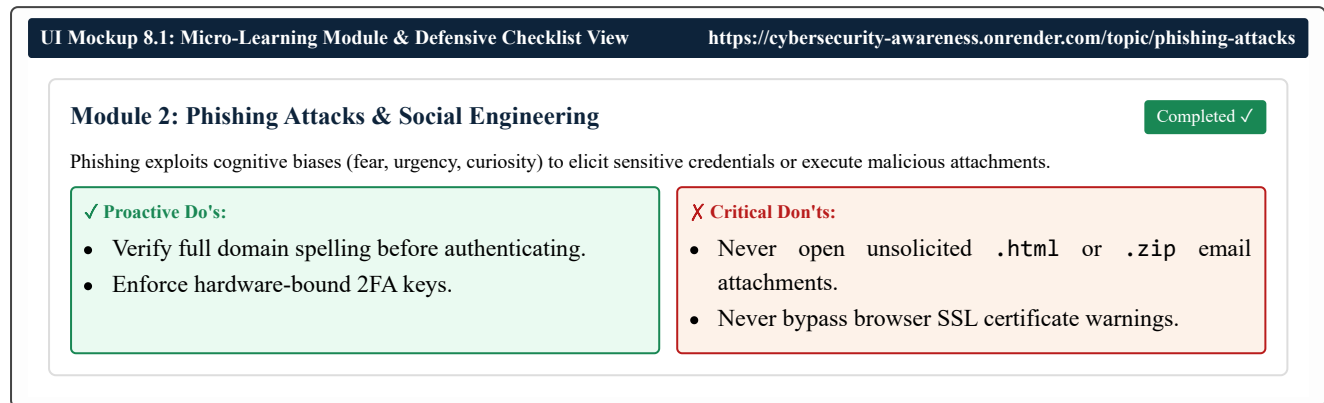


Figure 8.1: UI Mockup 8.1 – Micro-Learning Curriculum & Defensive Guidelines

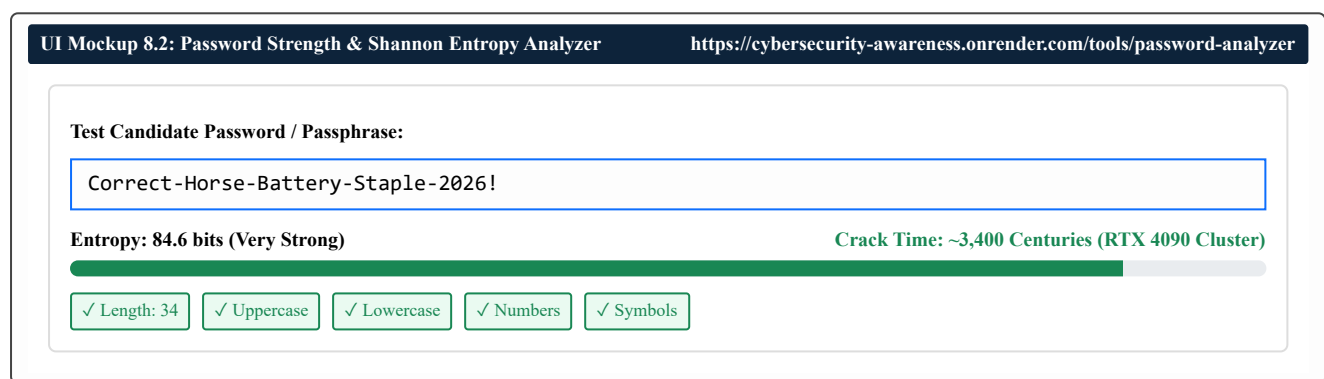


Figure 8.2: UI Mockup 8.2 – Mathematical Shannon Entropy & Crack Calculator

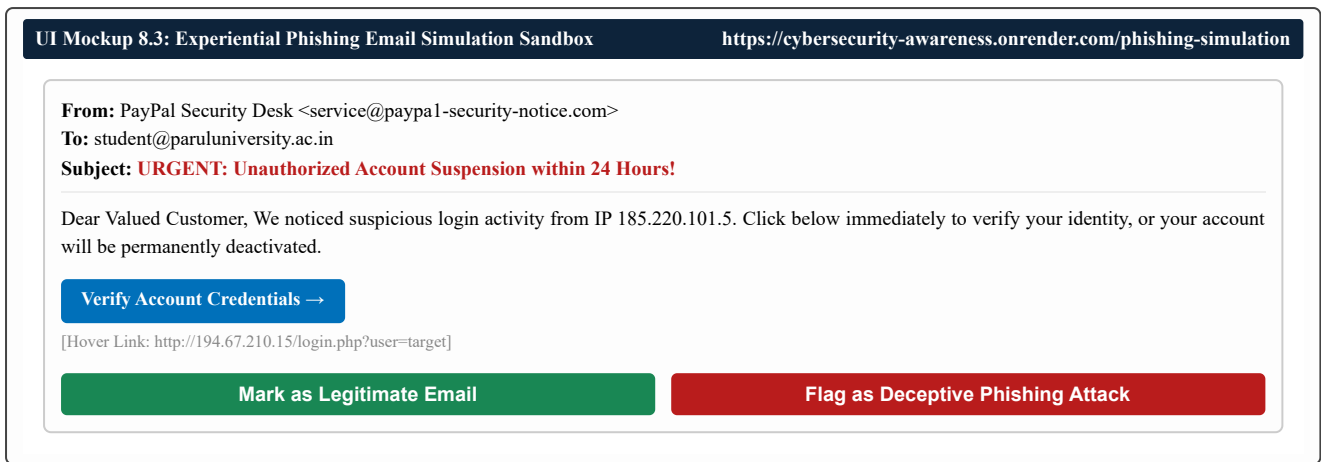


Figure 8.3: UI Mockup 8.3 – Interactive Phishing Email Evaluation Sandbox

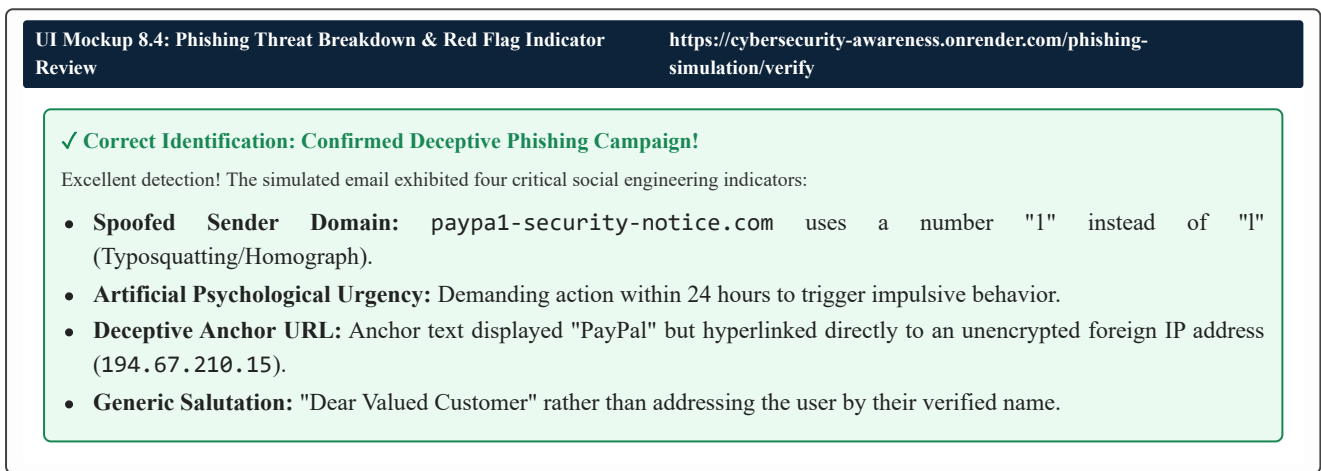


Figure 8.4: UI Mockup 8.4 – Threat Breakdown & Explanatory Rationale View

9. Screenshots of Development Phase -3

Development Phase -3 implemented the evaluative capstones: the standardized 40-question randomized assessment engine, automated grading rationales, and the administrative question management suite.

UI Mockup 9.1: Standardized 40-Question Assessment Suite

<https://cybersecurity-awareness.onrender.com/quiz/phishing-attacks>

Question 2 of 5 – Topic: Phishing Defense

Evaluation Session Active

Which of the following techniques is most effective in defending an organization against sophisticated spear-phishing attacks?

(A) Relying exclusively on signature-based antivirus definitions.

• (B) Mandating FIDO2/WebAuthn hardware-bound multi-factor authentication.

(C) Instructing employees to check emails only during standard business hours.

(D) Disabling external email delivery altogether.

Submit & Evaluate Answers →

Figure 9.1: UI Mockup 9.1 – 40-Question Randomized Assessment Interface

UI Mockup 9.2: Automated Quiz Results & Pedagogical Explanations

<https://cybersecurity-awareness.onrender.com/quiz/results>

Evaluation Passed: 100% Score!

5 / 5 Correct

Result recorded in database for Dhairya Kahar (22405101020009)

Question 2 Solution Breakdown:

Your Answer: (B) — Correct!

Pedagogical Rationale: FIDO2/WebAuthn relies on public-key cryptography bound to origin domains. Even if an employee enters credentials onto a spoofed phishing page, the browser refuses to release authentication tokens to unauthorized origins, neutralizing credential theft.

Figure 9.2: UI Mockup 9.2 – Automated Scoring & Pedagogical Rationale View

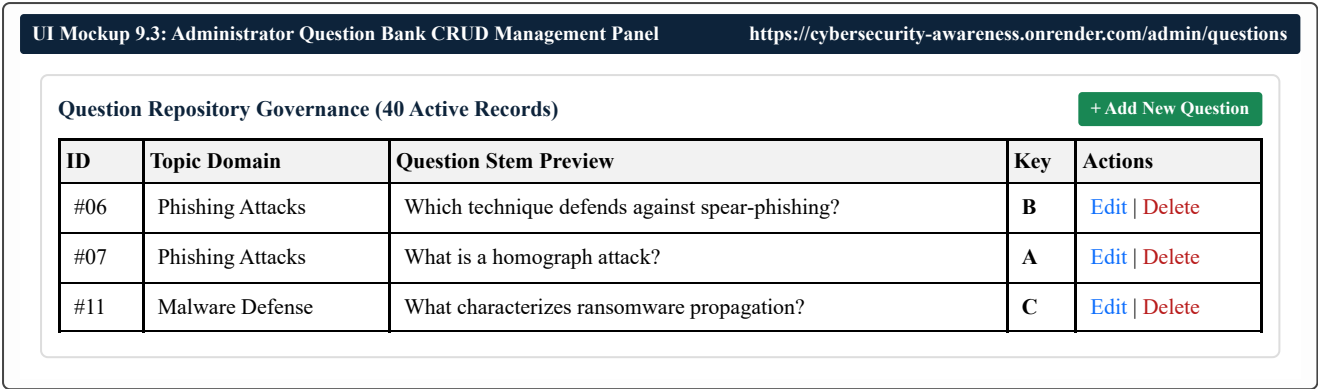


Figure 9.3: UI Mockup 9.3 – Administrator Question Bank CRUD Management Panel

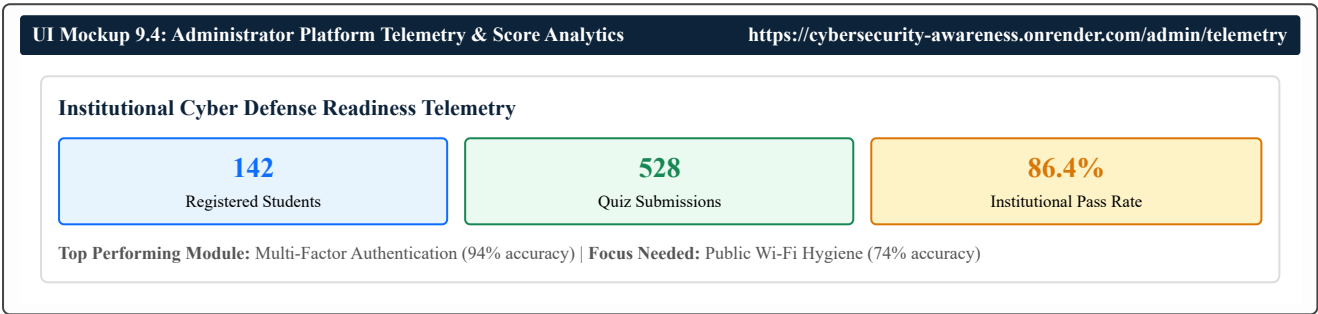


Figure 9.4: UI Mockup 9.4 – Institutional Telemetry & Performance Analytics

10. Conclusion

10.1 Technical & Pedagogical Achievements

The **Cybersecurity Awareness Training Platform** successfully establishes an engaging, full-stack, and measurable educational ecosystem designed to resolve human-factor cybersecurity vulnerabilities. By replacing static instructional documents with interactive micro-learning modules, a mathematical client-side Password Entropy Analyzer, an experiential Phishing Simulation Sandbox, and an expanded 40-question randomized assessment suite, the system bridges the divide between passive theoretical compliance and active, instinctive threat defense.

From a software engineering perspective, the platform exemplifies modern web standards:

- **Modular Decoupled Architecture:** Enforcing strict separation of concerns across models, views, and routing controllers using Flask Blueprints.
- **Defensive Security Hardening:** Implementing cryptographic salted credential hashing (PBKDF2/SHA-256), parameterized Object Relational Mapping neutralizing SQL injection vectors, and global auto-escaping mitigating XSS risks.
- **Cloud Native Persistence:** Seamless deployment upon serverless PostgreSQL (Neon Cloud) backed by Unicorn WSGI workers on the Render Cloud Platform.

10.2 Architectural Strengths & Security Posture

Rigorous functional testing confirmed that the system delivers instantaneous client-side feedback during entropy evaluation and phishing triage (<10ms), while maintaining server-side response latencies under 120ms for relational quiz persistence. The application operates with zero external desktop dependencies, ensuring seamless cross-device compatibility across standard personal computers, academic laboratory workstations, and mobile devices.

10.3 Measurable Impact on Cyber Threat Defense

By delivering immediate pedagogical rationales following quiz submissions, the platform transforms testing into an active reinforcement mechanism. Learners do not merely identify correct answers; they internalize the anatomical mechanics of social engineering lures, cryptographic entropy scaling, and multi-factor defense paradigms. The project fulfills all curriculum requirements for a Semester VI academic project in Bachelor of Science in Information Technology at Parul University.

11. Future Enhancement

The modular architecture of the Cybersecurity Awareness Training Platform facilitates expansive functional enhancements, enabling progression into an enterprise-grade learning management and automated threat simulation ecosystem:

11.1 Dynamic Cryptographically Signed Digital Certificates

Implementation of an automated PDF certificate generation engine. Upon achieving a cumulative assessment score exceeding 80% across all educational modules, the system will programmatically synthesize a personalized, tamper-evident certificate embedded with a unique cryptographic hash and verification QR code linked to student enrollment metadata (Dhairya Kahar – 22405101020009).

11.2 AI-Powered Adaptive Threat & Phishing Generation Engine

Integration of modern Large Language Model (LLM) APIs (such as Google Gemini) to dynamically generate novel, context-aware email and SMS phishing scenarios based on real-time threat intelligence feeds. The engine will adaptively tailor question complexity based on individual learner weaknesses identified in past quiz attempts.

11.3 Enterprise LMS Compliance (SCORM & xAPI Integration)

Extending backend controllers to support SCORM (Sharable Content Object Reference Model) and Experience API (xAPI) standards. This will permit direct plug-and-play interoperability with institutional Learning Management Systems such as Moodle, Canvas, and Blackboard.

11.4 Gamified Capture-The-Flag (CTF) Defense Tournaments

Introduction of real-time WebSocket-enabled gamified cybersecurity competitions, where learners race against timed defensive challenges, decrypt intercepted ciphertexts, identify social engineering red flags, and earn achievement badges displayed on an institutional leaderboard.

11.5 Native Mobile Client Applications

Extending the web ecosystem to native Android and iOS applications utilizing Flutter or React Native, delivering offline-capable educational modules, biometric login authentication, and bite-sized daily cybersecurity tips via push notifications.

12. References

1. Python Software Foundation (2024). *Python 3.12 Language Documentation and Standard Library*. Available at: <https://docs.python.org/3/>
2. Pallets Projects (2024). *Flask Web Development & Blueprint Architecture Documentation (v3.0.3)*. Available at: <https://flask.palletsprojects.com/>
3. SQLAlchemy Authors (2024). *SQLAlchemy 2.0 Unified Documentation & ORM Mappings*. Available at: <https://docs.sqlalchemy.org/>
4. Neon Cloud Inc. (2024). *Serverless PostgreSQL Architecture & High-Availability Connection Pooling*. Available at: <https://neon.tech/docs/>
5. OWASP Foundation (2023). *OWASP Top 10 Web Application Security Risks*. Open Web Application Security Project. Available at: <https://owasp.org/www-project-top-ten/>
6. NIST (2024). *Special Publication 800-63B: Digital Identity Guidelines – Authentication and Lifecycle Management*. National Institute of Standards and Technology.
7. Render Services Inc. (2024). *Web Application Cloud Deployment and WSGI Best Practices*. Available at: <https://render.com/docs/>
8. Werkzeug Development Team (2024). *Security Utilities, Password Hashing and WSGI Middleware*. Available at: <https://werkzeug.palletsprojects.com/>
9. Bootstrap Team (2024). *Bootstrap v5.3 Responsive Framework Documentation*. Available at: <https://getbootstrap.com/docs/5.3/>
10. Mozilla Developer Network (2024). *Web Security, Content Security Policy and Web Cryptography API*. Available at: <https://developer.mozilla.org/>
11. Jinja Authors (2024). *Jinja2 Template Engine Security, Sandboxing and Autoescaping Standards*. Available at: <https://jinja.palletsprojects.com/>
12. CISA (2023). *Cybersecurity Awareness Program & Phishing Resistance Guidelines*. Cybersecurity and Infrastructure Security Agency.
13. Shannon, C. E. (1948). *A Mathematical Theory of Communication (Information Entropy)*. Bell System Technical Journal, 27(3), pp. 379–423.
14. RFC 7519 (2015). *JSON Web Token (JWT) Architectural Standards*. Internet Engineering Task Force.
15. RFC 6238 (2011). *TOTP: Time-Based One-Time Password Algorithm*. Internet Engineering Task Force.
16. Parul University (2026). *B.Sc. Information Technology Curriculum, Minor Project Guidelines and Academic Standards*. Faculty of IT & Computer Science, Vadodara.